EE 3202.304

ECE Fundamentals Lab II

Lab 7: Operational Amplifiers

Name: Krish Dhage

Lab Partner: Vishal Manjunath

Dr. Lawrence Overzet

TA: Shideh Tayeb Naimi

11/03/2025

I. Experimental Results

Part 1 – Difference Amplifier

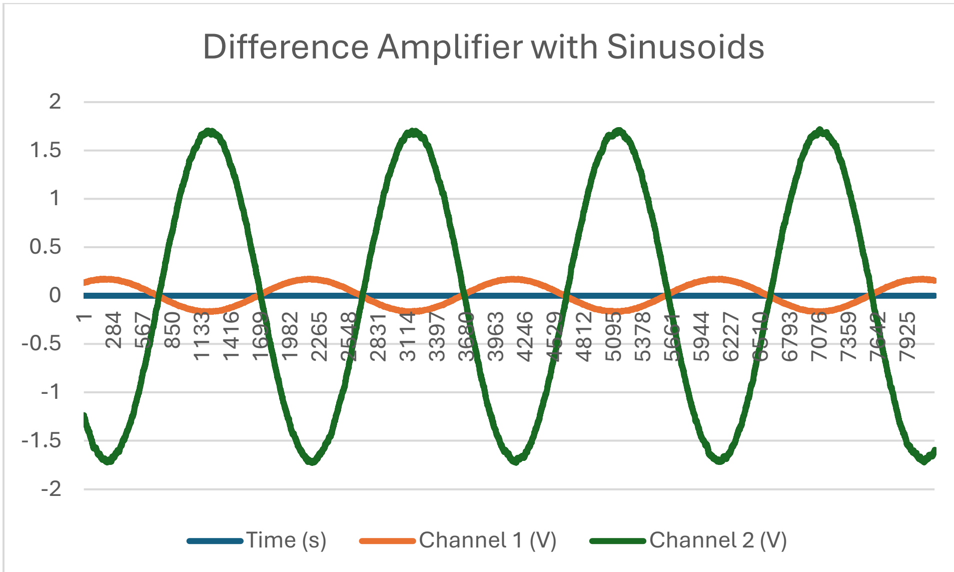

Figure 1.1: Difference Amplifier with two Sinusoid inputs

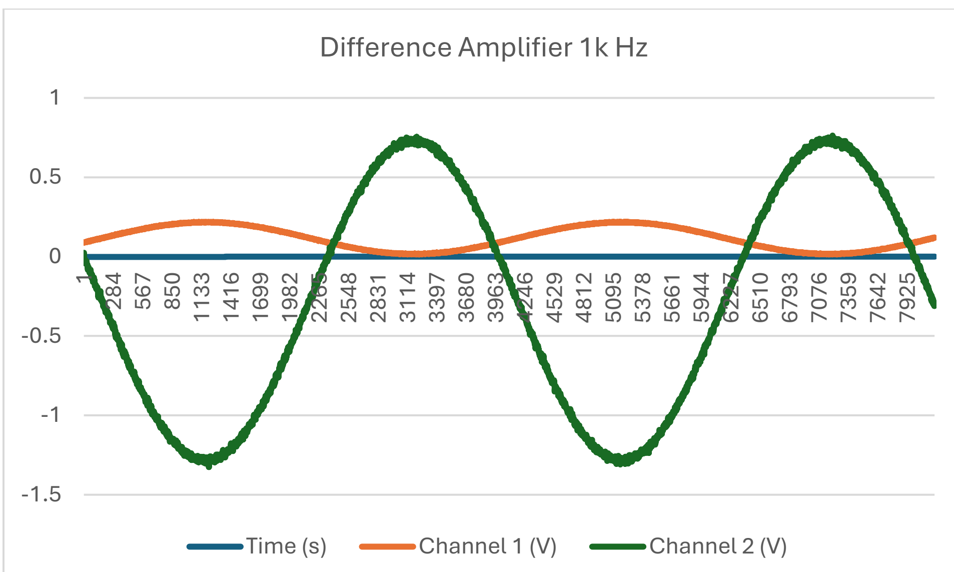

Figure 1.2: Difference Amplifier with 1k Hz

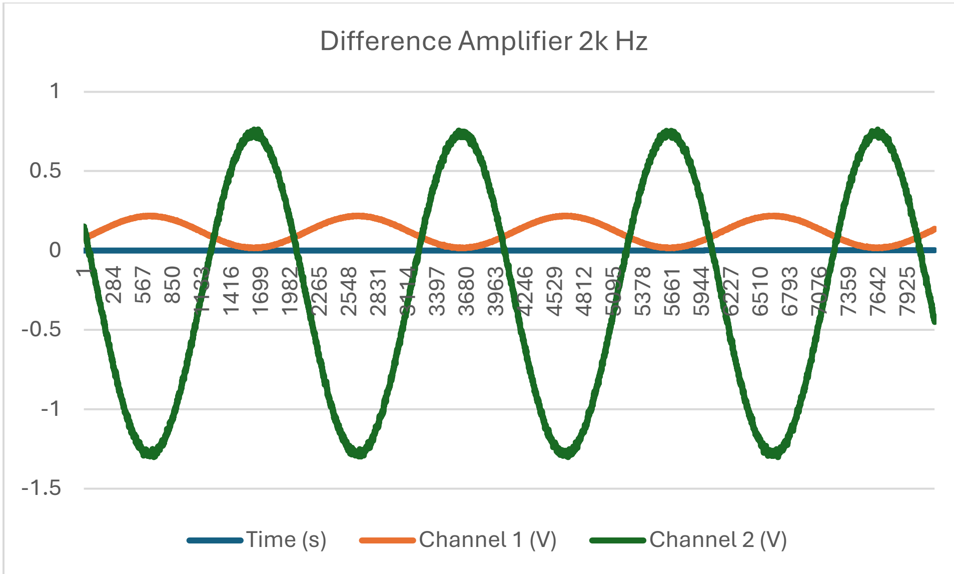

Figure 1.3: Difference Amplifier with 2k Hz

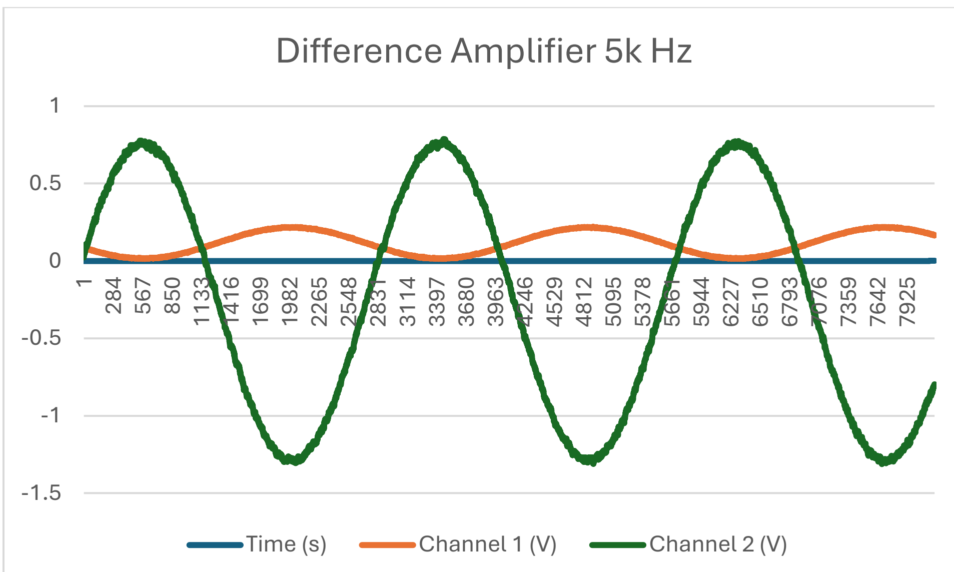

Figure 1.4: Difference Amplifier with 5k Hz

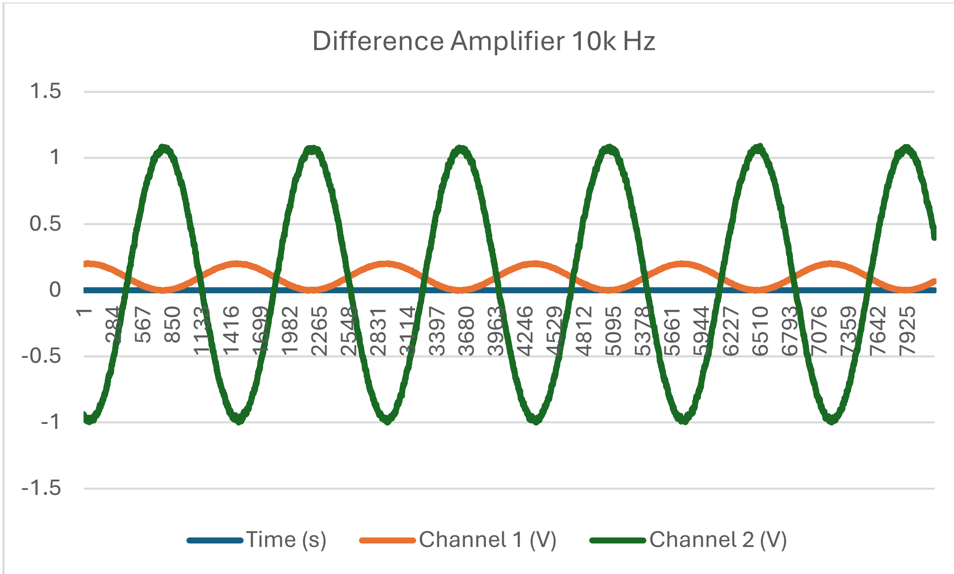

Figure 1.5: Difference Amplifier with 10k Hz

	Extent	Name
C2	Maximum	2.1630 V
C1	Maximum	106.83 mV

Figure 1.6: Sinusoid and pulse input

Part 2 – Differentiator

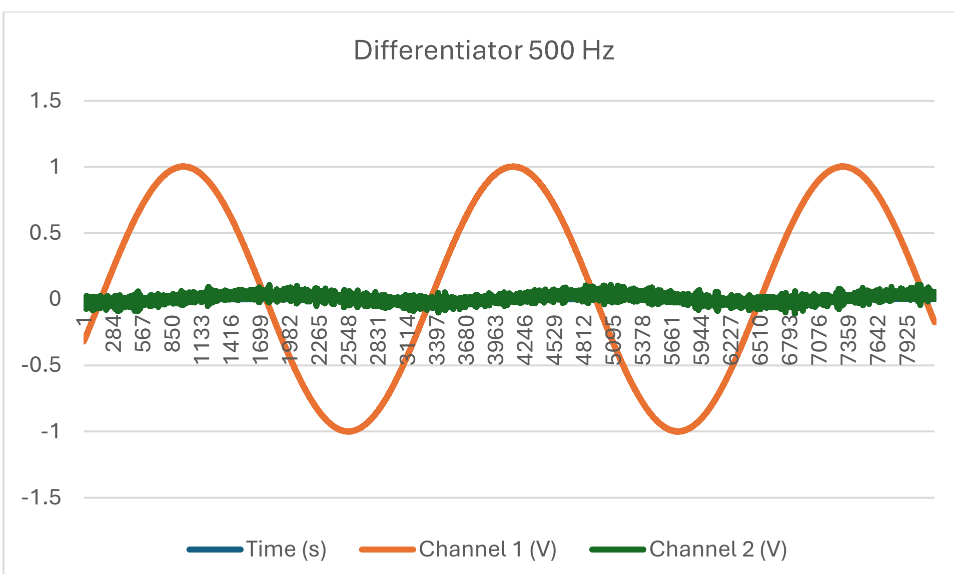

Figure 2.1: Differentiator for 500 Hz sine wave

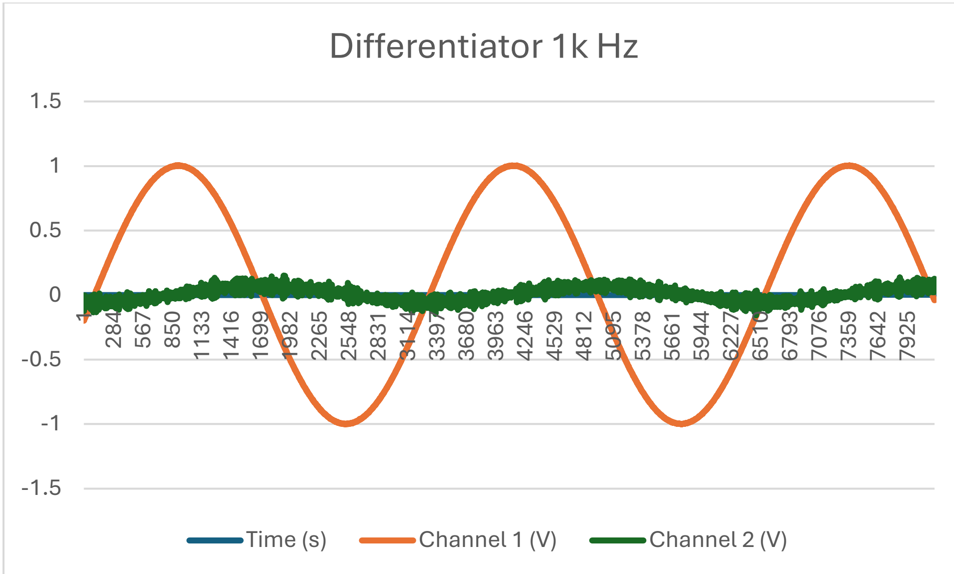

Figure 2.2: Differentiator for 1k Hz sine wave

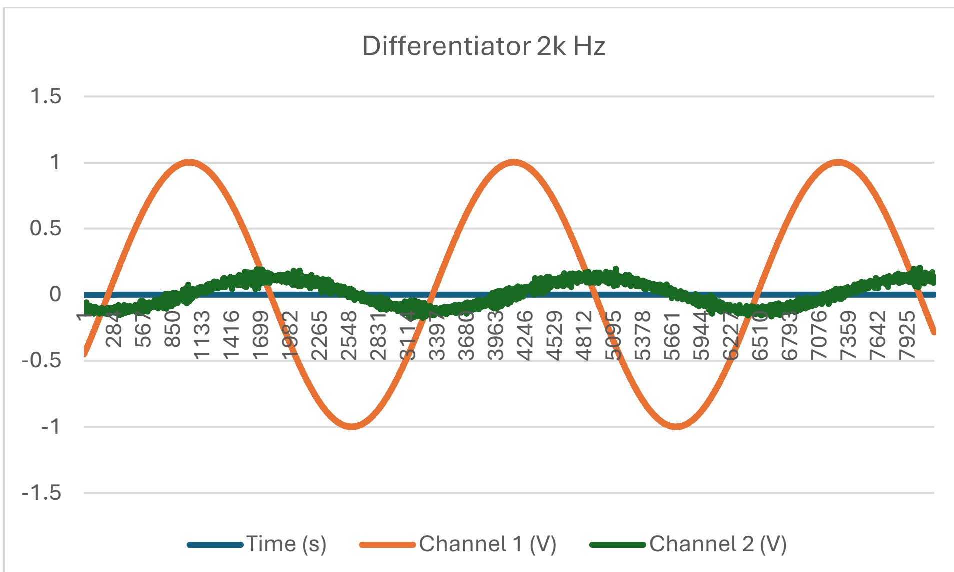

Figure 2.3: Differentiator for 2k Hz sine wave

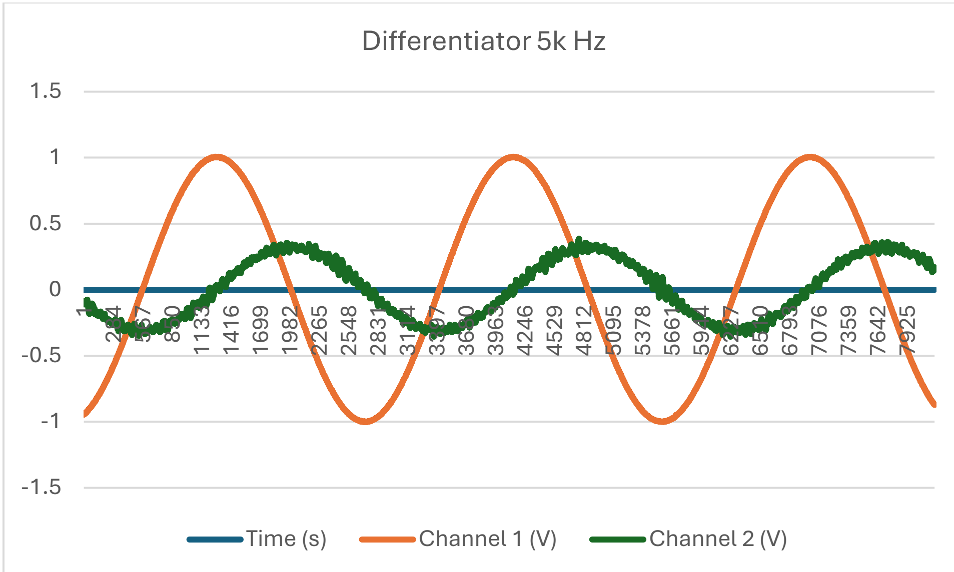

Figure 2.4: Differentiator for 5k Hz sine wave

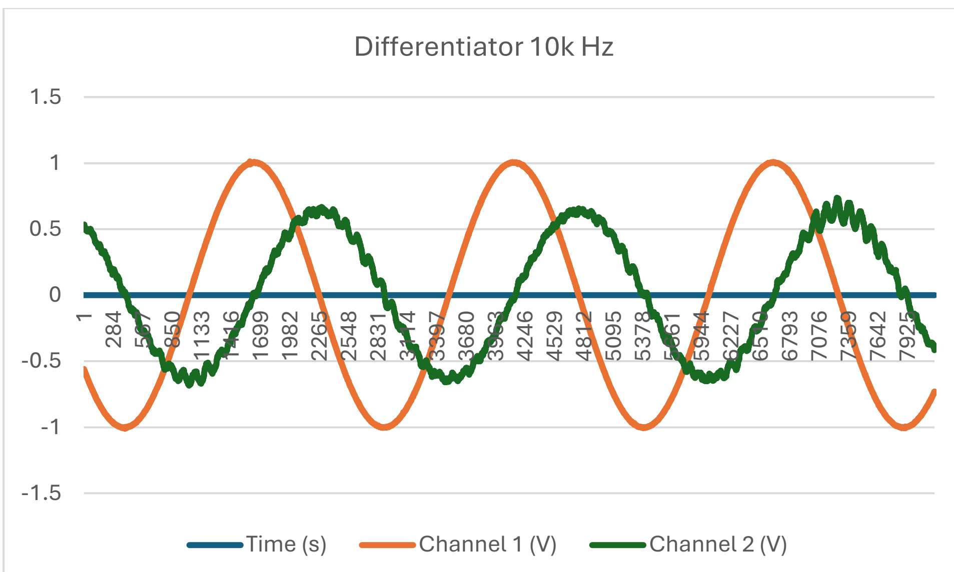

Figure 2.5: Differentiator for 10k Hz sine wave

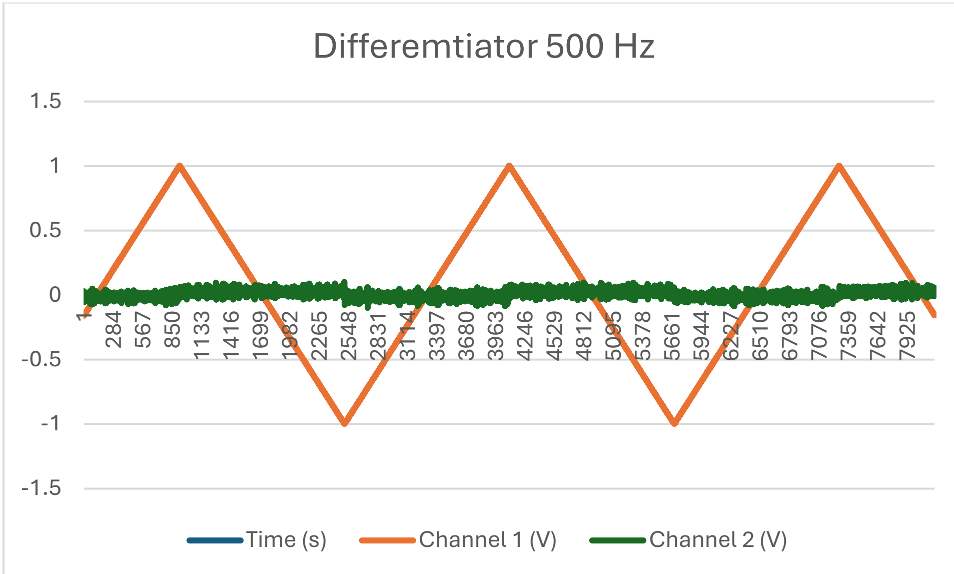

Figure 2.6: Differentiator for 500 Hz triangle wave

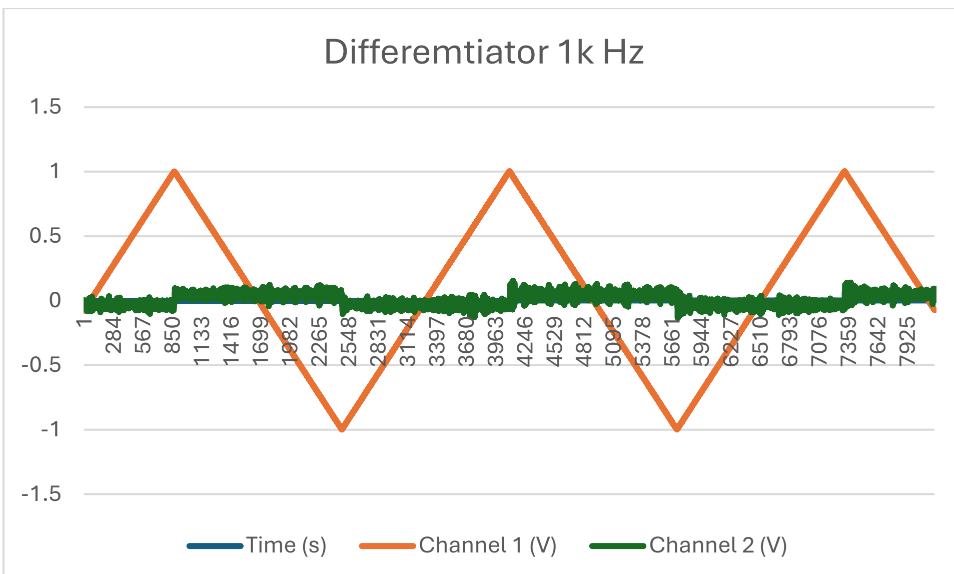

Figure 2.7: Differentiator for 1k Hz triangle wave

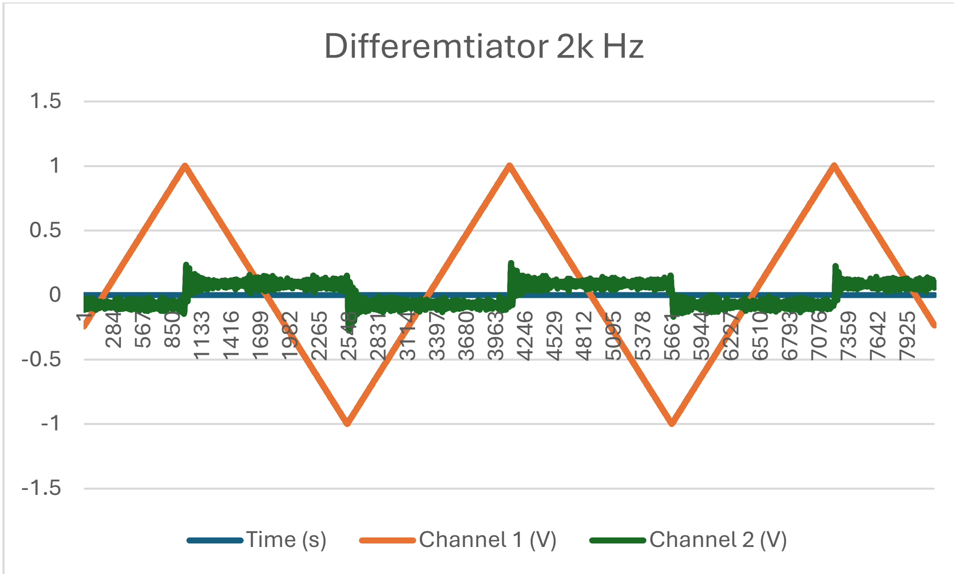

Figure 2.8: Differentiator for 2k Hz triangle wave

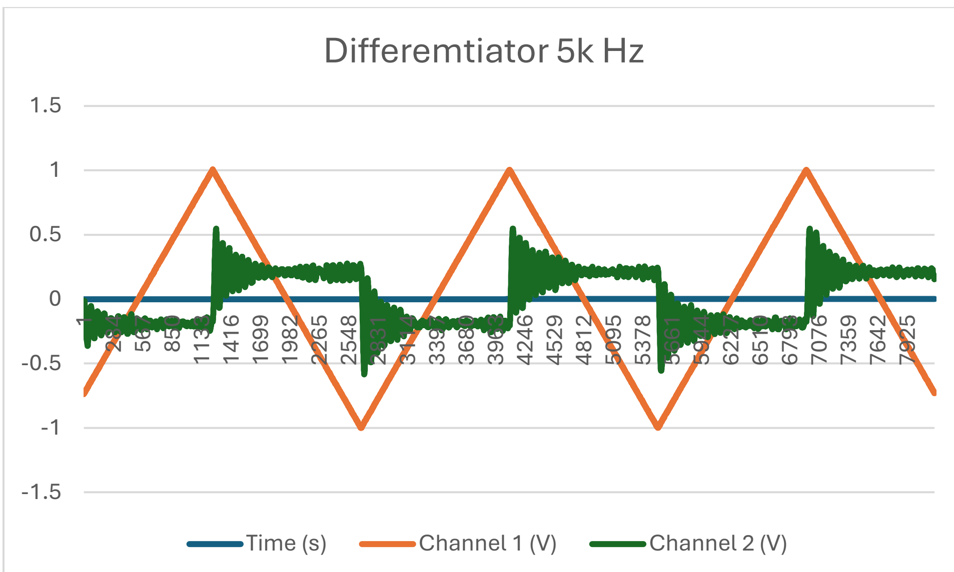

Figure 2.9: Differentiator for 5k Hz triangle wave

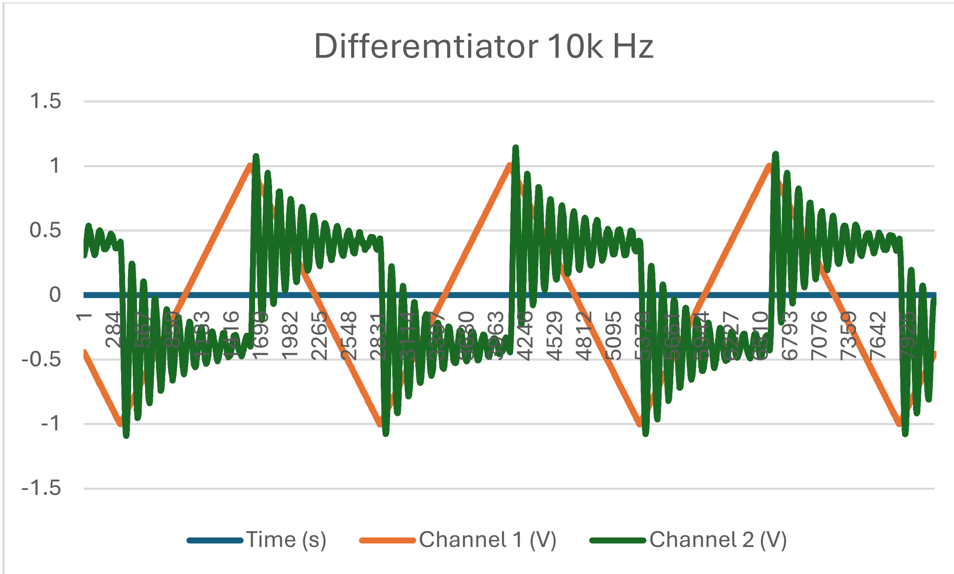

Figure 2.11: Differentiator for 10k Hz triangle wave

Part 3 – Integrator

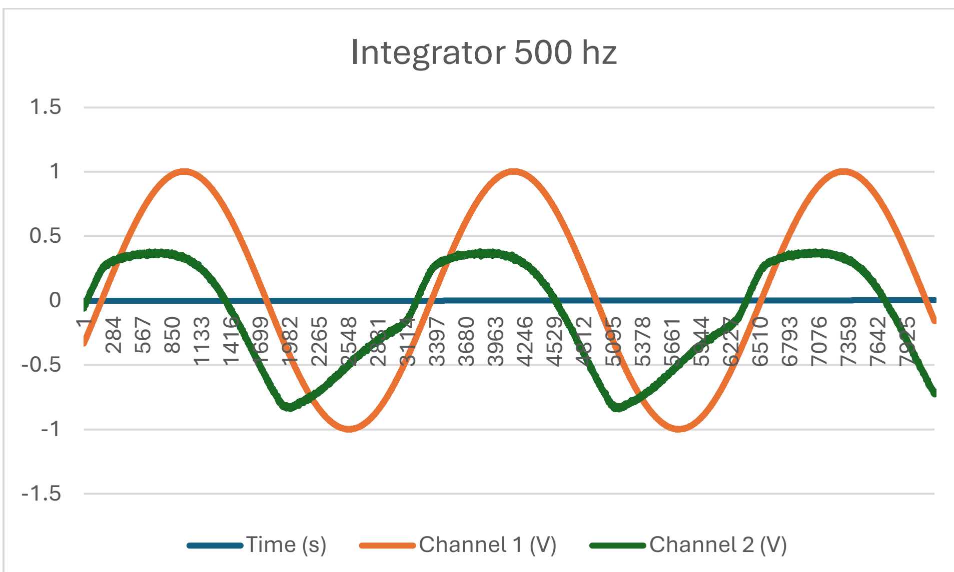

Figure 3.1: Integrator 500 Hz sine wave

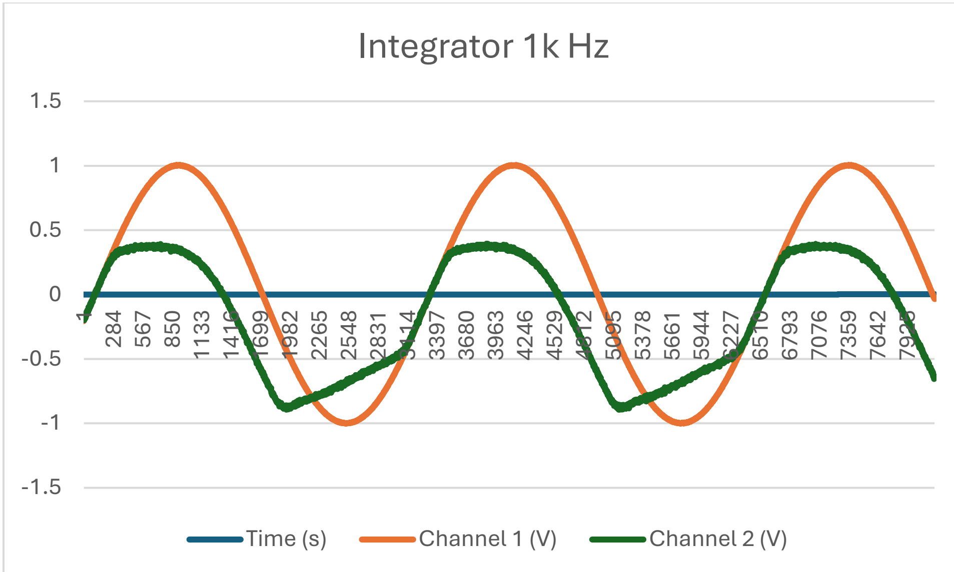

Figure 3.2: Integrator 1 k Hz sine wave

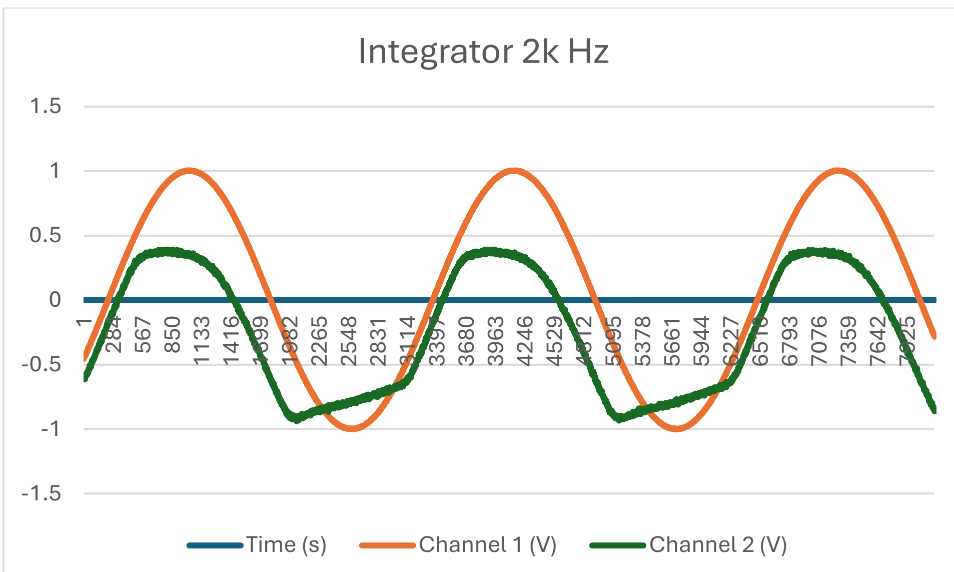

Figure 3.3: Integrator 2 k Hz sine wave

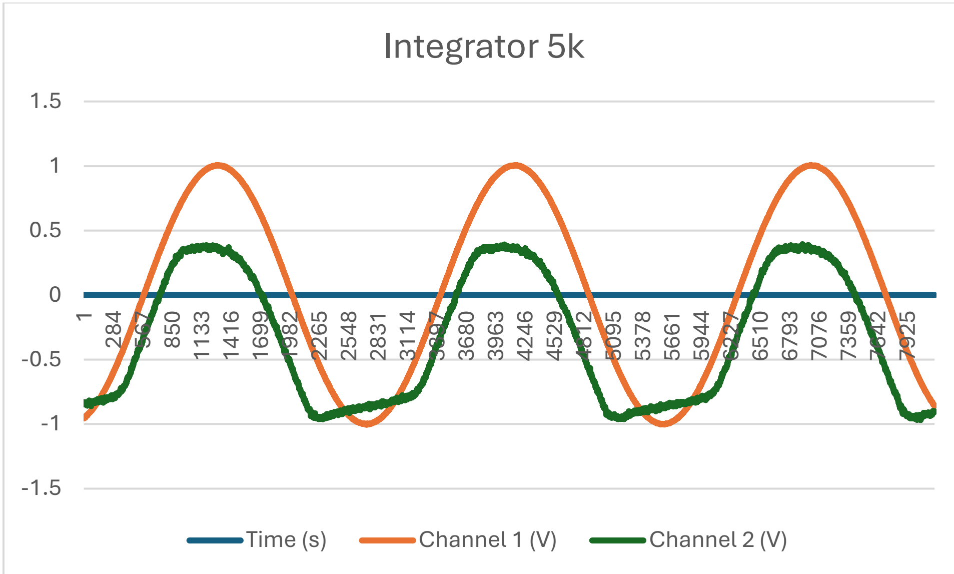

Figure 3.4: Integrator 5 k Hz sine wave

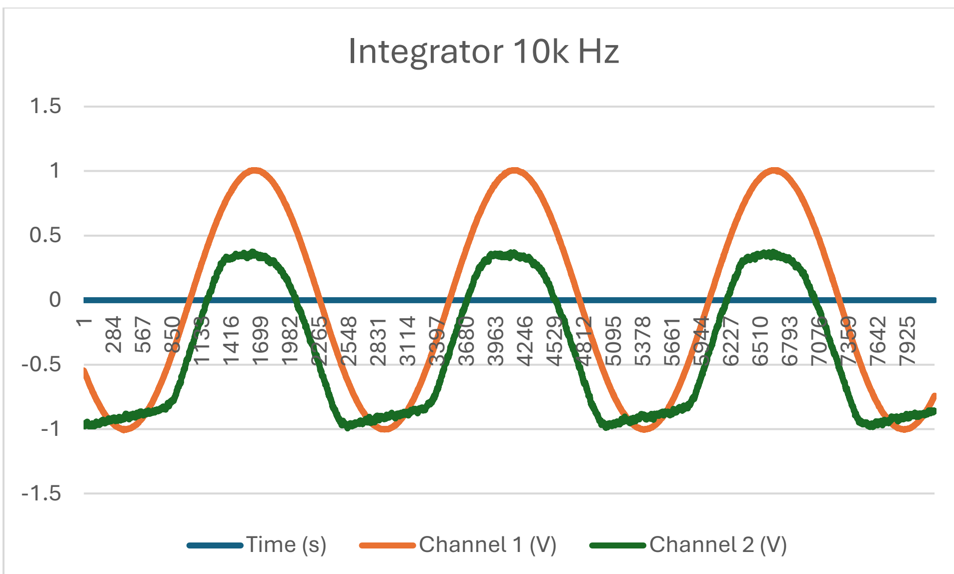

Figure 3.5: Integrator 10 k Hz sine wave

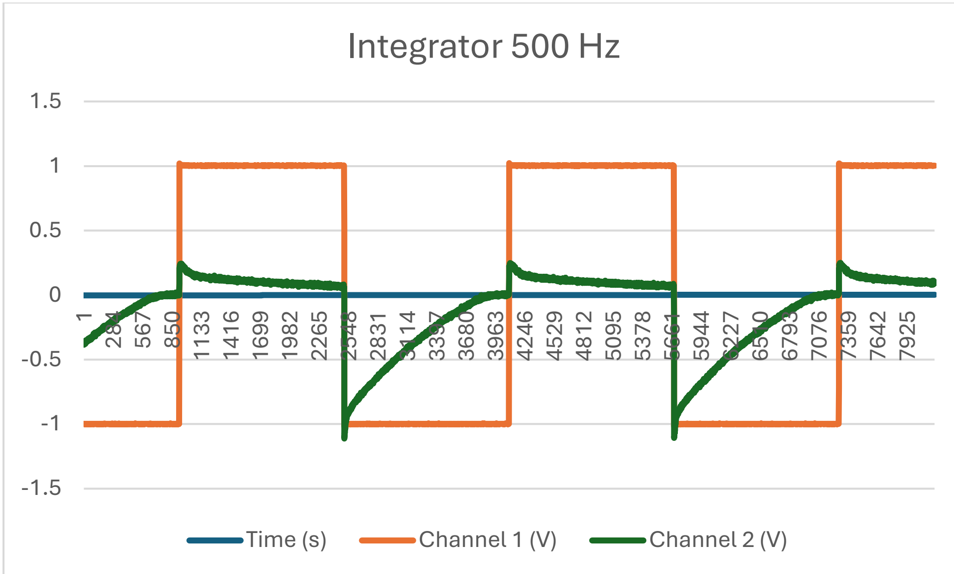

Figure 3.6: Integrator 500 Hz square wave

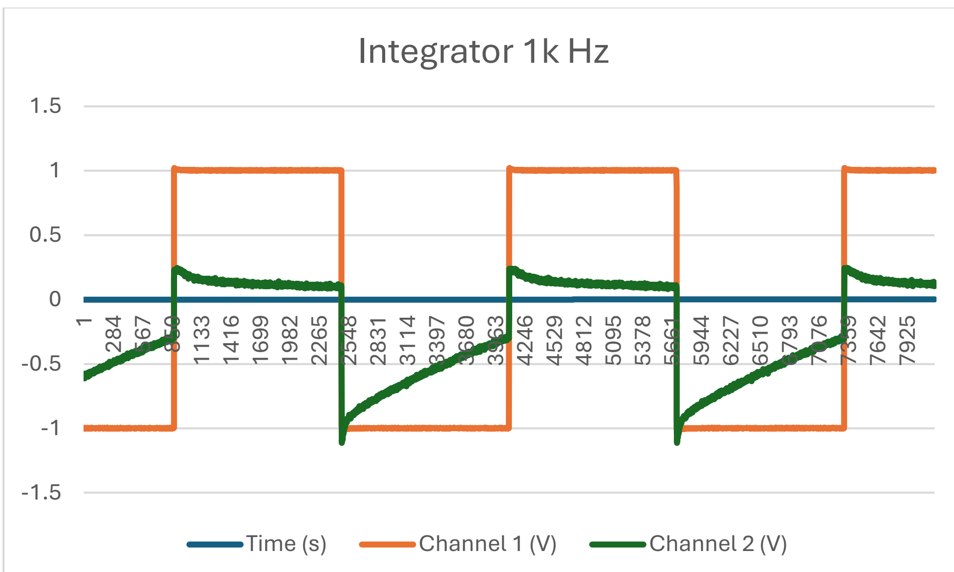

Figure 3.7: Integrator 1 k Hz square wave

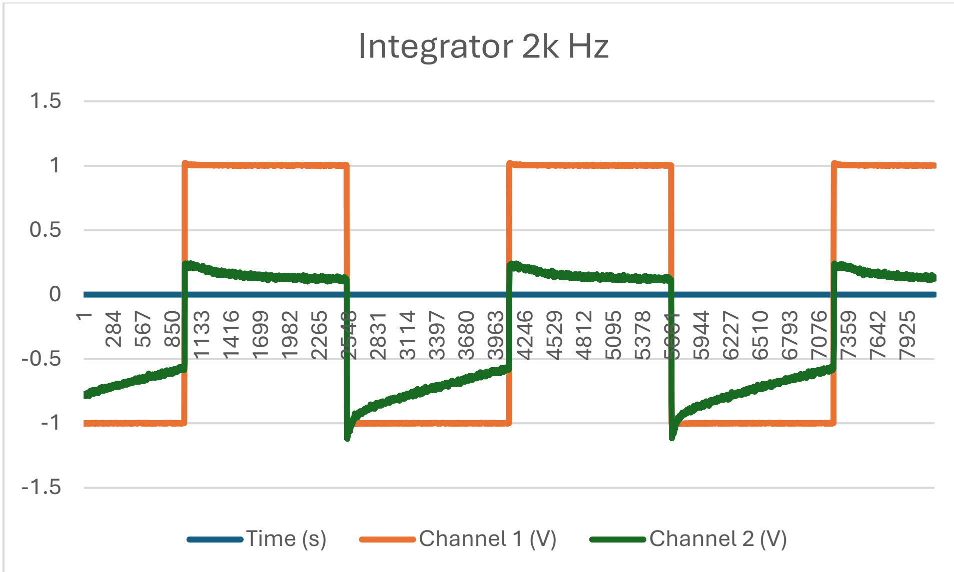

Figure 3.8: Integrator 2 k Hz square wave

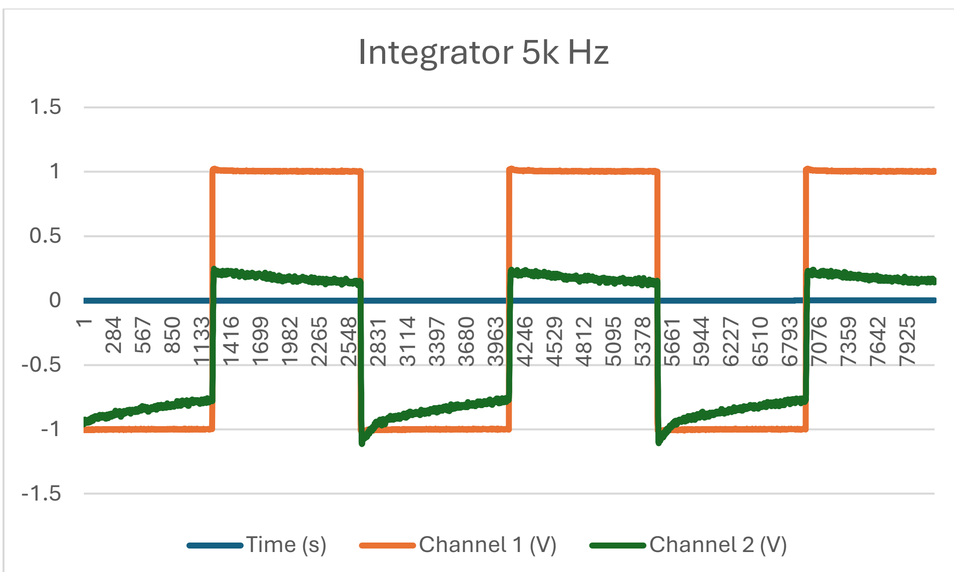

Figure 3.9: Integrator 5 k Hz square wave

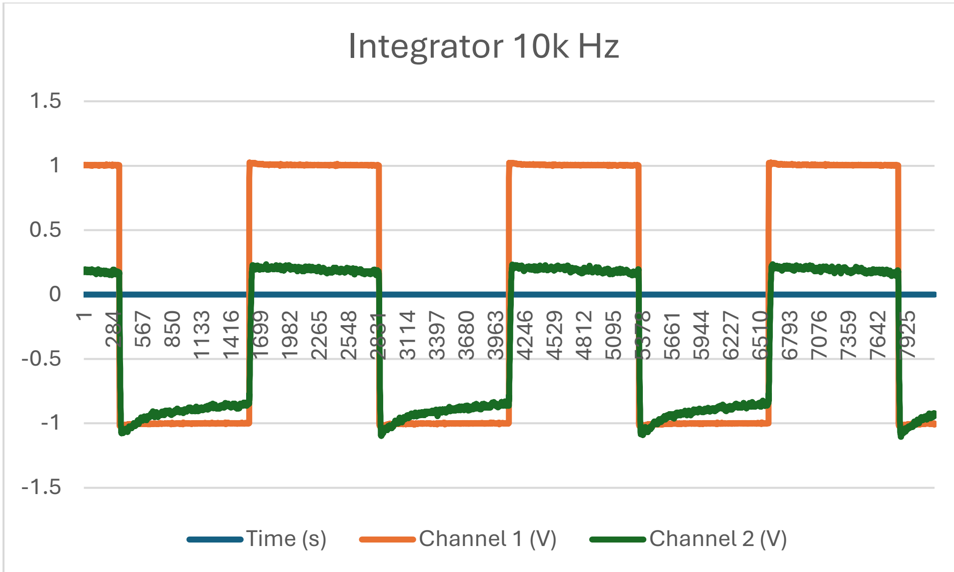

Figure 3.11: Integrator 10 k Hz square wave

Part 4 – Wien Bridge Oscillator

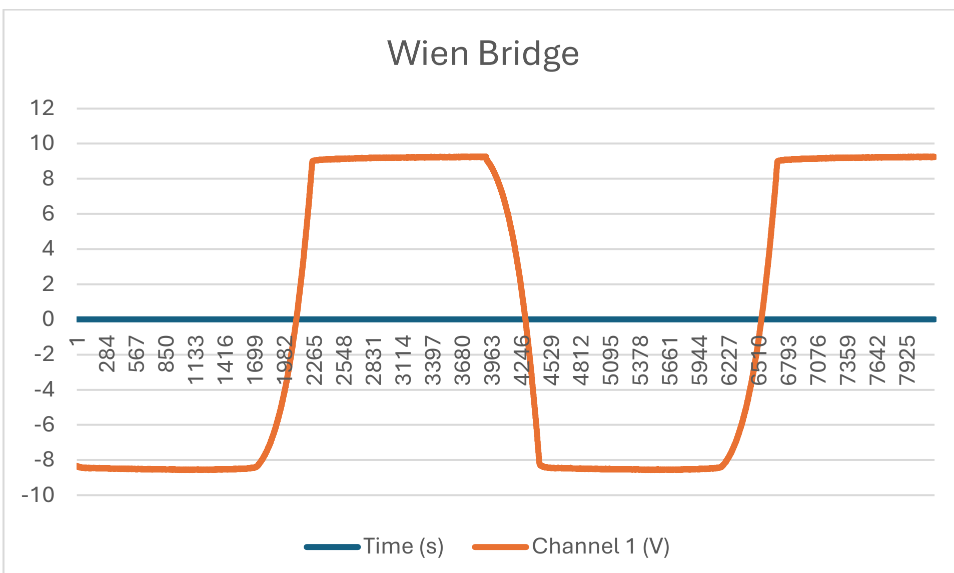

Figure 4.1: Wien Bridge Oscillator Output

II. Discussion and Analysis

In this experiment, several operational amplifier (op amp) circuits were constructed and analyzed to study their practical behavior compared to theoretical and simulated predictions. The TL082 and µA741 op amps were used throughout the lab, and their datasheets were consulted to ensure all tests were performed within safe operating limits for supply voltage, input, and output conditions. The objective was to explore the performance of key op amp configurations—specifically the difference amplifier, differentiator, integrator, and Wien bridge oscillator—and to observe how real-world imperfections influence their operation.

1. Difference Amplifier

The lab began with the construction of the difference amplifier that had been designed in the pre-lab assignment. The circuit was supplied with two sinusoidal inputs identical to those used in the simulation. The output waveform observed on the oscilloscope matched the expected differential signal, showing that the amplifier properly amplified the voltage difference between the two input signals. Minor discrepancies, such as small amplitude variations and phase shifts, were attributed to resistor tolerances and the inherent limitations of the physical components compared to the ideal conditions assumed in simulation.

The second test used a different pair of input signals: a sinusoidal waveform with a peak-to-peak amplitude of 100 mV at 5 kHz and a pulse waveform with a 200 mV peak-to-peak amplitude, 10% duty cycle, 90° phase shift, and a frequency of 10 kHz. Theoretically, the output should represent an amplified difference between these two signals. Experimentally, the resulting waveform displayed a sinusoidal base with distinct pulse distortions, confirming the proper subtraction behavior of the circuit. Small deviations in amplitude and waveform sharpness were observed, most likely caused by the op amp's limited slew rate and bandwidth. Overall, the difference amplifier demonstrated consistent and predictable differential amplification, validating the design.

2. Differentiator Circuit

The differentiator circuit was then built using the TL082 op amp, following the configuration developed in the pre-lab. The circuit was tested with both sinusoidal and triangular input waveforms across a range of frequencies. When a sinusoidal input was applied, the differentiator produced a cosine-like output waveform, which demonstrated that the output voltage is proportional to the rate of change of the input signal. As the frequency increased, the amplitude of the output also increased, confirming the expected frequency-dependent gain characteristic of a differentiator.

To further verify the differentiating action, a triangle wave was used as the input signal. The differentiator responded by producing a square wave output, since the derivative of a linear ramp (the triangle wave) is a constant voltage. This provided clear visual confirmation of the differentiator's intended function. At higher frequencies, the output began to show more noise and distortion, caused by the op amp's sensitivity to high-frequency components and limitations in bandwidth. Nonetheless, the differentiator circuit performed as predicted, accurately demonstrating the theoretical relationship between input and output waveforms.

3. Integrator Circuit

Next, the integrator circuit was constructed using the µA741 op amp. The same frequencies used in the pre-lab simulations were tested to examine the circuit's frequency response. When a sinusoidal signal was applied to the input, the output waveform appeared phase-shifted and of reduced amplitude, resembling a cosine waveform, consistent with the theoretical integral relationship between the two. The amplitude of the output decreased as the input frequency increased, illustrating the inverse frequency dependence of the integrator's gain.

The integrator was also tested using a pulse waveform input to observe its behavior with non-sinusoidal signals. As expected, the output transformed the sharp pulses into gradually rising and falling slopes, resulting in a triangular output waveform. This occurred because the integrator effectively accumulates the voltage of the pulse over time, converting each pulse's area into a corresponding change in voltage. Some small DC drift and offset were visible in the output, which are typical in real integrator circuits due to capacitor leakage and bias currents. Overall, both tests confirmed that the circuit behaved as an ideal integrator, producing output signals that matched the predicted waveform shapes.

4. Wien Bridge Oscillator

Finally, the Wien Bridge oscillator circuit was built as designed in the pre-lab. Upon powering the circuit, oscillations were observed on the oscilloscope, confirming that the feedback network sustained self-generated waveforms. The output waveform was not a perfect sine wave but was periodic and stable, allowing the oscillation frequency to be measured. The measured frequency closely matched the theoretical frequency calculated from the resistor and capacitor values used in the feedback network. The amplitude of oscillation stabilized over time, limited by the op amp's internal gain and supply voltage range. This experiment effectively demonstrated the principle of positive feedback and the use of an op amp to generate continuous oscillations without an external input.

Summary

Each circuit demonstrated consistent behavior with theoretical predictions and simulation results, confirming the fundamental principles of op amp applications. The difference amplifier successfully subtracted input signals; the differentiator accurately produced rate-of-change outputs, including transforming triangle waves into square waves; the integrator correctly generated cumulative outputs, converting pulse inputs into smooth triangular signals; and the Wien Bridge oscillator produced a stable, self-sustaining waveform. Minor discrepancies between simulation and experiment were primarily due to real-world non-idealities such as component tolerances, parasitic capacitances, and op amp limitations. Overall, the lab provided a clear understanding of how op amps can be configured for a wide range of signal processing and waveform generation tasks.